A chronic peptic ulcer is a classic example of **chronic active inflammation**: the background is chronic (repeated cycles of injury and healing), with acute inflammation superimposed at the ulcer base each time the injury flares.

1. How the ulcer forms: imbalance between damage and defense

An ulcer develops when aggressive factors overwhelm mucosal protective factors.

- **Aggressive factors:** gastric acid and pepsin, *H. pylori*, NSAIDs, bile reflux, smoking, and hypersecretory states such as Zollinger-Ellison syndrome.
- **Protective factors:** mucus-bicarbonate barrier, prostaglandins (PGE2), epithelial renewal, good mucosal blood flow, and tight junctions.

H. pylori (about 70-90% of duodenal ulcers, about 70% of gastric ulcers):

- Urease splits urea into ammonia, which buffers acid around the bacterium.
- Flagella and adhesins let it colonize the mucus layer. Virulence factors (CagA, VacA) damage epithelial cells and trigger IL-8 release.
- Antral gastritis lowers somatostatin and raises gastrin, which increases acid secretion. This is the main mechanism in duodenal ulcers, where acid exposure causes gastric metaplasia of the duodenum that *H. pylori* can then colonize.
- In the gastric body, atrophic gastritis can lower acid output, and the ulcer results from mucosal weakness rather than hyperacidity.

NSAIDs inhibit COX-1, which reduces prostaglandins. This lowers mucus and bicarbonate, reduces blood flow, and impairs epithelial repair.

2. Why inflammation is both acute and chronic

- **Acute component:** Injury attracts **neutrophils** (driven by IL-8 and other chemokines), which release enzymes and reactive oxygen species. This produces the active necrotic, exudative layer at the base.
- **Chronic component:** Persistent injury, especially from *H. pylori*, brings **lymphocytes and plasma cells** into the surrounding mucosa. Healing attempts form granulation tissue and fibrosis, and the ulcer cycles between healing and breakdown over months to years.

3. Histology of a chronic ulcer (four zones, from lumen to depth)

1. **Necrotic fibrinoid debris** with neutrophils (acute inflammation)
2. **Nonspecific inflammatory infiltrate** of neutrophils, lymphocytes, and plasma cells

3. **Granulation tissue** with new capillaries, fibroblasts, and macrophages (active repair)
4. **Fibrous scar** that may extend into the muscularis propria and adjacent tissue (chronicity)

4. Complications from the process

- **Bleeding:** erosion of a vessel (e.g., gastroduodenal artery in posterior duodenal ulcers)
- **Perforation:** through the wall, especially anterior duodenal ulcers
- **Penetration** into adjacent organs such as the pancreas
- **Gastric outlet obstruction:** from scarring and edema

Key point: Duodenal ulcers are almost never malignant, while gastric ulcers can be, so gastric ulcers are biopsied.